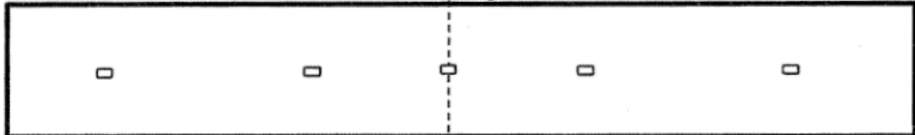


6. (a) (i)	$\Delta y = \frac{\lambda D}{a}$ $\frac{(4.0-0) \times 10^{-2}}{10} = \frac{\lambda(1.8)}{0.3 \times 10^{-3}}$ $\lambda = 6.666667 \times 10^{-7} \text{ m}$ $\approx 6.67 \times 10^{-7} \text{ m or } 667 \text{ nm}$	1M+1M	1A	3	Explanation in terms of formula $\lambda = \Delta y \frac{a}{D}$ is NOT accepted as a is the slit separation given, which is fixed.
(ii) To ensure that the light through the 2 slits diffracts enough to interfere / overlap.	1A 1A	2			
(b) (i)	$d \sin \theta = m\lambda$ $\frac{10^{-3}}{500} \sin \theta = 6.67 \times 10^{-7} \text{ m}$ $\theta = 19.471221^\circ \approx 19.47^\circ$	1M	1M 1A	3	
Separation between central and first-order bright spot $= 1.8 \tan 19.47^\circ$ $= 0.636396 \text{ m} \approx 0.636 \text{ m}$					
(ii)	centre of the pattern 	1A 1A	2		
symmetry about the central bright spot (2 nd order shown) separation between 2 nd and 1 st -order bright spots is larger					